

NATIONAL CHEMISTRY CONTEST 2023

“Every aspect of the world today – even politics and international relations – is affected by chemistry.”

Linus Pauling

Name(s): _____ Last name: _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Exam: **Day 1. ELEVENTH GRADE**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 1 problem set, with an individual exam for 11th grade. The exam has 4 problems (problem 1, 5 items; problem 2, 3 items; problem 3, 4 items; problem 6, 11 items) and a total of 8 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.
- Number the lower part of the exam, each page and indicate the number of total answer pages, for example, “1 of 7”, “2 of 7” and so on.
- You are only given four and a half hours, control timing carefully: pay attention that the exam has a large number of questions. Do not waste time on unnecessary answers!

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas law	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zeroth in the Celsius's scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst' equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{C_{\text{red}}}{C_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
Standard conditions	$p^\circ = 100 \text{ kPa} = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	Van't Hoff i factor	$i = 1 + \alpha \cdot (v - 1)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	Cryoscopy	$\Delta T_c = i \cdot K_c \cdot b$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mmHg} = 101,325 \text{ kPa}$			
$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$			
$1 \text{ dm}^3 = 1 \text{ L} = 1000 \text{ mL}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$M \equiv \text{mol/L}$			

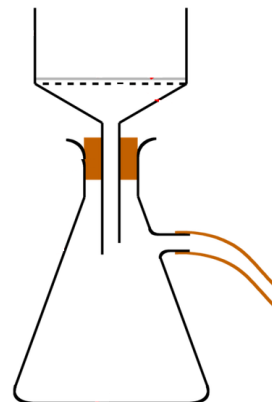
1																18	
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 1. Identification of substances (common for all grades)

1.1	1.2	1.3	1.4	1.5	Total	Total
11	11	8	3	7	40	10%

A concentrated aqueous solution (problem solution) of a blue colored salt **1** was prepared. Using pH paper, it was determined that the solution was practically neutral (neither acid nor basic). The coloration of the solution on the flame was blue-green. Next, a few milliliters of the problem solution were taken and placed in several test tubes, arranged in a rack. A concentrated sodium hydroxide solution was added to the first test tube and the appearance of a “sky-blue” (light blue) precipitate was observed. When excess sodium hydroxide was added, the precipitate dissolved, resulting in a deep blue solution. In the second test tube, excess concentrated hydrochloric acid was added and the solution resulting from the mixture had a green-yellow color. A solution of concentrated barium chloride was added to the third test tube and the formation of a white precipitate was observed. A solution of concentrated calcium chloride was added to the fourth test tube and the formation of a white precipitate was observed.



1.1. Indicate the cation and anion present in salt **1**. Write all the equations (ionic or molecular) of the chemical reactions that allowed their identification.

Finally, a portion of the test solution was concentrated by heating; subsequently, it was cooled and blue crystals of salt **1** were obtained. The solid was separated by vacuum filtration, placed in a porcelain dish, on top of a tripod and heated slightly with a Bunsen burner. The capsule was held with pincers and carefully stirred with a glass shaker until all the solid was observed to turn into a greyish-white powder.

1.2. About the experimental procedures, answer:

1.2.a) What does the change on the color of the solid during heating indicate?

1.2.b) Indicate in the vacuum filtration apparatus represented in the figure: Büchner funnel, Kitasato collecting flask, rubber section, filter paper and perforated rubber stopper.

1.2.c) State the function of the tools: rack, test tube and spoon spatula.

1.2.d) Mention four protection tools used in the chemistry laboratory.

The concentrated aqueous solution of a colorless salt **2** was subjected to an analysis similar to that described above. When adding hydrobromic acid, a light-yellow precipitate was obtained, which dissolved in concentrated ammonia and no gas evolution was observed. The addition of sodium sulfide caused a black precipitate to form and an increase in the acidity of the solution. When the pH of the original solution was increased by adding sodium hydroxide, the appearance of a brown precipitate was observed, which dissolved in concentrated ammonia. It was also observed that the solution of salt **2** turned dark colored while time passed.

1.3. Indicate the cation present in salt **2**. Write all the equations (ionic or molecular) of the chemical reactions that allowed their identification.

1.4. About the experimental procedures, answer:

1.4.a) Why do solutions of **2** turn black while they age?

1.4.b) Which of the following anions could be present in salt? Cl^- , NO_3^- , HCO_3^- , CO_3^{2-} , CH_3CO_2^- , SO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$, I^- , SO_3^{2-} , PO_4^{3-} , $\text{S}_2\text{O}_3^{2-}$, $\text{C}_2\text{O}_4^{2-}$.

The yellow, concentrated aqueous solution of a salt **3** changed to an orange coloration when acidified with concentrated sulfuric acid. More acid was added to a portion of the orange solution until a deep red needle-

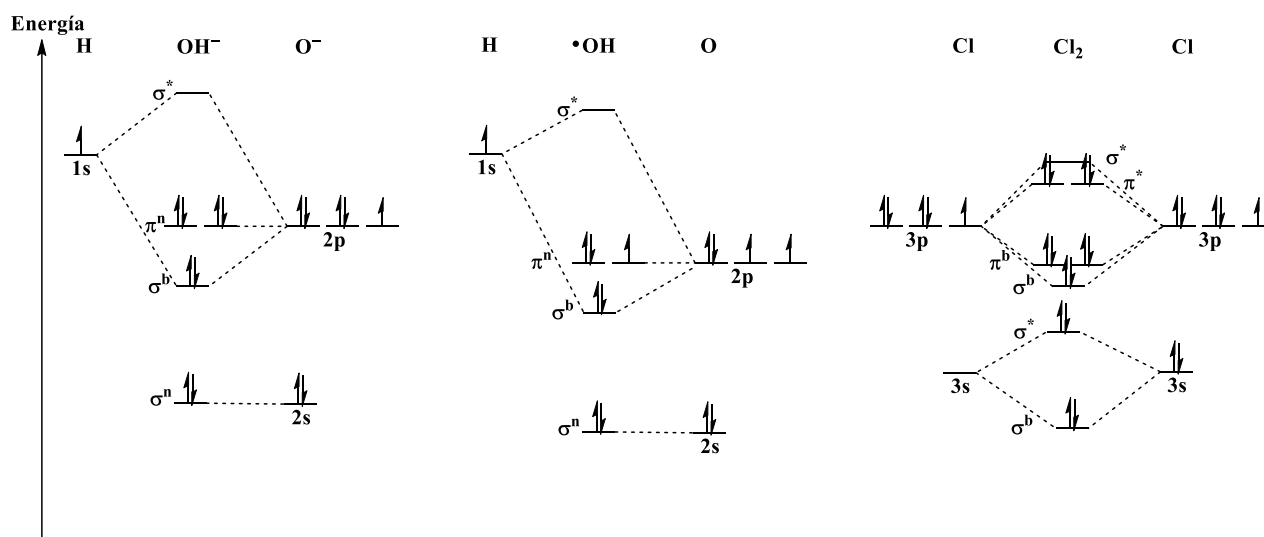
shaped precipitate was obtained, the product of a non-redox reaction. Hydrogen peroxide was added to a portion of the orange solution, the solution turned green after some time and released a colorless, odorless gas. Finally, when the solutions of salts **2** and **3** were mixed, an intense red precipitate was obtained.

1.5. Indicate the anion present in salt **3**. Write all the equations (ionic or molecular) of the reactions that allowed their identification.

Problem 2. Reactivity (only 10th and 11th grades)

2.1	2.2	2.3	Total	Total
24	7	9	40	10%

The figure illustrates the molecular orbitals (MO) diagrams for the hydroxide anion, the hydroxyl radical and the chlorine molecule. In addition, you have additional information.



Transition state ($\ddagger$): Particular, unstable configuration of the atoms during a reaction that has the maximum energy. When the atoms of the reactants “abandon” this intermediate state in the reaction path (coordinate), the formation of products will always occur. The activation energy is the difference between the energy of the transition state and the energy of the reactants.

MO: Molecular orbital, AO: atomic orbital, FO: frontier orbital, HOMO: Highest energy occupied molecular orbital, LUMO: Lowest energy unoccupied molecular orbital, SOMO: Molecular orbital occupied by a single electron, electrophile or Lewis acid: electron density deficient species, nucleophile or Lewis base: electron density rich species.

2.1. Complete the following statements as briefly as possible.

2.1. Complete the following statements as briefly as possible.

2.1.a) The O atom attracts with _____ strength the _____ electrons when compared to the Cl atom. Therefore, the energy of the OAs of O is lower than the energy of the OAs of Cl and, consequently, the O atom is _____ electronegative than the Cl atom.

2.1.b) Species OH^- and Cl_2 are _____ while radical OH is _____ an intense magnetic field, because the first ones have all their electrons _____, while the last one _____.

2.1.c) The bond order in all three species is equal to _____.

2.1.d) In Cl_2 the magnitude of the interaction (overlap) between the AOs is _____ in comparison with the species OH^- and $\cdot\text{OH}$. This is because the atomic radius of the chlorine atom is _____.

2.1.e) The energy of the MOs in the anion OH^- is _____ than the energy of the MOs in the radical $\cdot\text{OH}$ because _____.

2.1.f) The frontier orbitals of these species are: HOMO π^n and _____ for the anion OH^- , _____ and _____ for the radical $\cdot\text{OH}$, and _____ and _____ for the Cl_2 molecule.

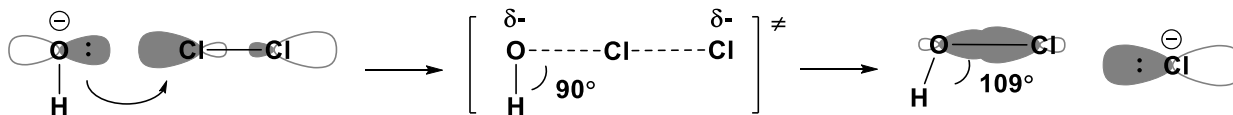
2.1.g) Partial charge takes a positive, zero, and negative value on the atoms _____, _____ and _____, respectively.

2.1.h) Anion OH^- in its reactions behaves as _____ because it has a high electron density. On the contrary, radical $\cdot\text{OH}$ and Cl_2 molecule behave as _____.

2.1.i) Anion OH^- and radical $\cdot\text{OH}$ are very reactive species because the first one has _____, while the second one _____.

2.1.j) In all cases, the number of MOs that are formed are numerically _____ to the total number of AOs of the atoms originating the species.

The reaction mechanism between the OH^- anion and the Cl_2 molecule is represented in the following scheme, where the FOs involved in the reaction, the direction of approach of the reactants, the geometries of the transition state and of the final product and the flow of electrons are highlighted.



2.2. Complete the following statements as briefly as possible.

2.2.a) Anion OH^- donates the electron pair occupying the filled HOMO π^n to the empty _____ of Cl_2 molecule.

2.2.b) O-H bond _____ during the reaction because the FO of OH^- that participates is non-bonding, while the Cl-Cl bond _____ during the reaction because the FO of Cl_2 that participates is _____.

2.2.c) When an additional OH^- ion reacts with hypochlorous acid, anion _____ and a neutral molecule of _____ are obtained. During this reaction, OH^- _____ the electron pair occupying the filled HOMO π^n to the empty LUMO $\sigma^*(2\text{sp}^3\text{O}-1\text{sH})$ of HClO .

2.3. Draw a scheme similar to the one shown above for the reaction that occurs between two $\cdot\text{OH}$ radicals forming hydrogen peroxide. Represent the FOs involved in the reaction, the direction of approach of the reactants, the geometries of the transition state and the final product, and the electron flow (by arrows).

Problem 3. Gold alloys (common for all grades)

3.1.	3.2	3.3	3.4	Total	Total
6	9	16	9	40	10%

Gold is a non-reactive metal used since ancient times in coins, jewelry and as a symbol of power. Its metallic luster is characteristic and is not lost in the presence of other metals. For this reason, it is common that those who trade gold try to deceive their clients with gold alloys with different metals. The jeweler classifies the samples that contain gold in carats (22, 18 or 12 K) depending on the mass content of gold present (equal to or greater than 91,7%, 75% or 50%, respectively).

A laboratory worker was given 100 mg of different samples containing gold for analysis, obtaining the results below. All the experiments were carried out at 0 °C and 101.325 kPa.

Calculate the number of carats (K) each gold sample contains. Formulate the balanced equations of all the chemical reactions that occur and find what other metal forms the pink gold alloy.

3.1. Sample of purple gold, containing aluminum: When dissolved in excess hydrochloric acid it generates 30.0 mL of a colorless gas. $V_m = 22,4$ L/mol.

3.2. Sample of pink gold, which contains a metal X: It was dissolved stoichiometrically in 36.5 mL of a solution of nitric acid ($M = 63,02$ g/mol) of mass concentration 5,00 g/L, leaving a dry residue of 54, 0 mg unreacted.

3.3. Sample of white gold, which contains platinum: It was dissolved in excess of aqua regia (HNO_3/HCl) until completely dissolved, generating 35,0 mL of reddish-brown NO_2 . It is known that when dissolving the sample two soluble products are formed whose percentage compositions by mass are Au : Cl : H equal to 58,0% : 41,7% : 0,3% and Pt : Cl : H equal to 47,6% : 51,9% : 0,5%. Find the formulas of the species formed.

3.4. Sample of blue gold, which contains iron: The sample was immersed in 25,0 g of a 4,00% mass silver nitrate solution until the iron completely reacted, which oxidized to the 2+ state. The treated gold sample was dried and weighed, resulting in a mass of 172 mg. The remaining silver nitrate solution was reacted with excess sodium chloride. The silver chloride solid after drying had a mass of 714 mg.

Problem 6. Alcohol oxidation (only 11th and 12th grade)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	6.10	6.11	Total	Total
4	18	6	4	4	5	5	13	10	6	5	80	20%

In a laboratory, 0,10 M ethanol and 0,75 M potassium dichromate solutions at pH 3,5 are available. To determine the kinetic parameters of the previous reaction, three 100 mL volumetric flasks were prepared by adding a different volume of the ethanol solution in each one. Then they were completed up to the mark with the potassium dichromate solution until three solutions of 100 mL were obtained in each case. A beam of light was passed through a spectrophotometer at a wavelength where only dichromate absorbs, and the absorbance was measured at the beginning of the mixture and 20 s later. All experiments were carried out at 25 °C. Using the molar absorptivity of dichromate, it was possible to measure the initial speed of the reaction, obtaining the data that is summarized in the following table:

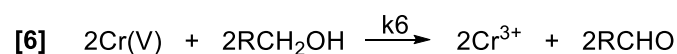
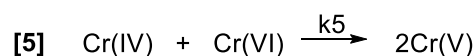
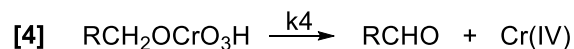
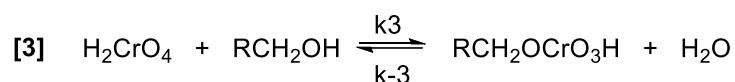
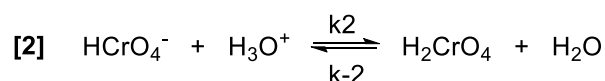
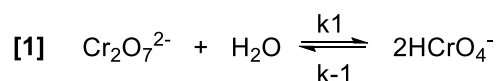
Volumetric flask 100 mL	Volume of RCH ₂ OH (mL)	Initial rate (M/s)
1	50,0	$2,401 \cdot 10^{-9}$
2	60,0	$2,577 \cdot 10^{-9}$
3	70,0	$2,604 \cdot 10^{-9}$

Assume that after forming the three solutions the pH is approximately 4.5. The rate law of the reaction has the general formula $v_r = k_r [\text{RCH}_2\text{OH}]^x [\text{Cr}_2\text{O}_7^{2-}]^y [\text{H}_3\text{O}^+]^z$, where $z = 1$.

6.2. Calculate the partial orders of reaction (x , y) and the rate constant using the method of initial rates and the data in the table. To do this, calculate the concentrations of alcohol and dichromate in each flask after the solutions have been mixed.

Note: If you cannot solve the previous part, consider that $k_r = 0,00250 \text{ u}$.

This reaction has been widely studied and it is known that it occurs through a mechanism that consists of six steps as shown below:

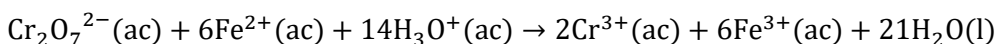


6.3. If equation [4] is the slow step and equilibria [1], [2], and [3] are fast, find an expression for the rate law assuming that these equilibria hold during the reaction. Express the total rate constant in terms of the rate constants for each step.

6.4. The reaction under study follows the Arrhenius equation. Briefly explain what are the terms activation energy and pre-exponential factor that appear in this equation.

6.5. When heating the previous solutions up to 33 °C, the rate constant of the experimentally determined reaction increased by 2,78 times. Calculate the activation energy and the pre-exponential factor.

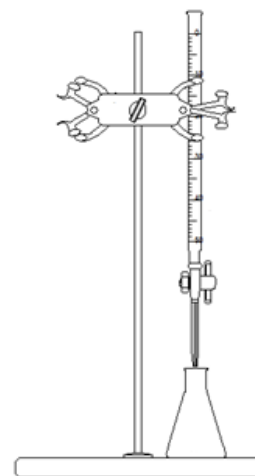
The determination of reaction orders can also be performed using a redox titration. Unreacted potassium dichromate with alcohol can be determined with a standard solution of iron(II) sulfate. To carry out the study, a 100 mL volumetric flask (problem solution) was prepared where 60 mL of 1,40 M ethanol and 20,0 mL of 0,100 M potassium dichromate were added and the total volume was completed with a 3,0 M solution for H₂SO₄. Then, samples of 10,0 mL of the problem solution were taken every 30 s and titrated with an acidified standard solution of 0,0900 M Fe²⁺, using 1,1-phenanthroline as redox indicator. Before beginning each titration, a small amount of phosphoric acid was added to avoid color interference from the Fe³⁺ ions that form.



6.6. Indicate in the figure the laboratory tools used in a titration.

6.7. Calculate the initial pH of the problem solution. $\text{pK}_{a2}(\text{H}_2\text{SO}_4) = 1,99$.

Note: If you cannot solve the previous item, consider pH = 0,20 to answer the next question.



6.8. Estimate the value of the potential at the equivalence point of the first titration ($t = 0$ s, see the table with the volume of solution consumed). $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0,77$ V, $E^\circ(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) = +1,33$ V.

Under these conditions, when the concentrations of alcohol and hydronium ions are much higher than the concentration of dichromate, it can be assumed that the reaction rate depends only on the concentration of the latter, since the other concentrations remain practically constant and then $v_r = k_r' \cdot [\text{Cr}_2\text{O}_7^{2-}]^y$, where $k_r' = k_r \cdot [\text{RCH}_2\text{OH}]^x [\text{H}_3\text{O}^+]^z$. Mathematically, it can be shown that depending on the value of the partial reaction order of the dichromate (y) the following equations stand

Order	$y = 1$	$y \neq 1$
Rate law	$\ln \left(\frac{[\text{Cr}_2\text{O}_7^{2-}]}{c_0(\text{Cr}_2\text{O}_7^{2-})} \right) = -k_r' \cdot t$	$[\text{Cr}_2\text{O}_7^{2-}]^{(1-y)} = -(1-y) \cdot k_r' \cdot t + c_0(\text{Cr}_2\text{O}_7^{2-})^{(1-y)}$

The volume of iron(II) sulfate needed to reach the equivalence point in each case was

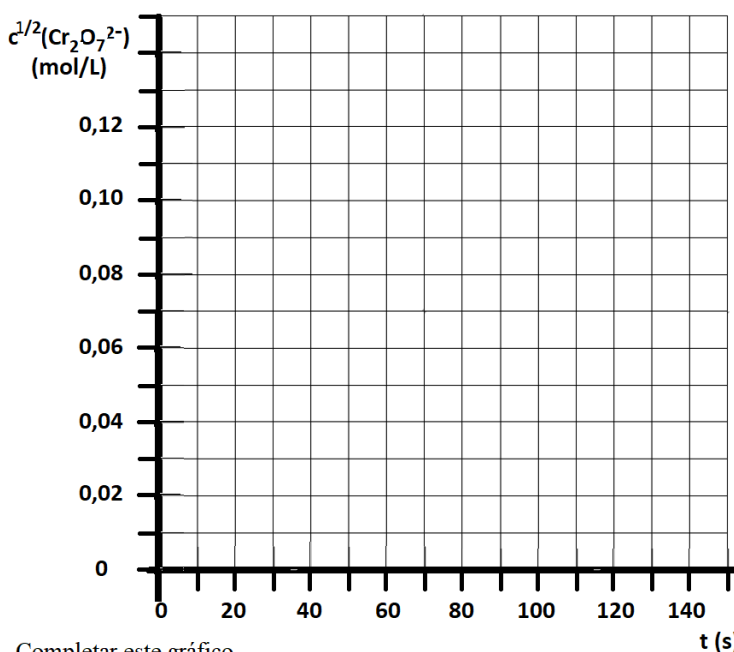
Time (s)	0	30	60	90	120	150
Volume of Fe^{2+} (mL)	13,33	9,76	6,75	4,29	2,39	1,04

6.9. Calculate the concentration of dichromate at each time and complete the graph to show that the partial reaction order is $y = \frac{1}{2}$ in this case.

6.10. Using the graph you constructed or using a linear regression calculation, calculate the value of the rate constant k_r' and the partial reaction order x corresponding to the alcohol. Use the values of k_r and $[\text{H}_3\text{O}^+]$ you calculated in **6.2** and **6.7**, respectively.

The determination of dichromate can also be carried out indirectly by iodometry, using starch as indicator. A fraction of the problem solution can be reacted with excess iodide and the triiodide ion formed is then titrated with a thiosulfate solution of known concentration.

6.11. Write balanced ionic equations for all reactions that occur during the determination of dichromate by iodometry.



Completar este gráfico.